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MILITARY SPECIFICATION

CAMERA SET, STILL PICTURE
KS-7( )

This specification is approved for use by USA Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one view-type, 8-inch by 10-inch (203mm x 254mm), copying camera designated as Camera Set, Still Picture KS-7( ) (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MMM-A-1617	Adhesive, Rubber Base, General Purpose
PPP-B-20	Bag, Cotton, Mailing
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard
PPP-B-1055	Barrier Material, Waterproofed, Flexible
PPP-C-843	Cushioning Material, Cellulosic
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant (For Carton Sealing)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-B-117	Bag, Sleeve and Tubing-Interior Packaging
MIL-B-131	Barrier Materials, Watervapor-proof, Flexible, Heat-Sealable
MIL-H-3526	Holders, Photographic Sheet Film, Double Lock Rib
MIL-G-12803	Gasket Material, Non-Metallic
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-C-55442	Cables Assemblies and Cord Assemblies Packaging of

STANDARDS

FEDERAL

FED STD No 356	Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-150	Photographic Lenses
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Components. The major components of Camera, Set, Still Picture KS-7( ) are as follows:

Camera, Still Picture, Copying (see 3.3)
Lenses, Camera, Copying (see 3.4)
Stand, Camera (see 3.7)
Holders, Film (see 3.8)
Holder, Filter (see 3.9)
Lighting System (see 3.10)

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3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.3 Camera, Still Picture, Copying.

3.3.1 Camera body. The camera body sections shall be fabricated of light weight metal alloy, or other suitable material, to provide sufficient strength, rigidity and freedom from vibration to maintain all operating parts and the optical system of the camera in proper alignment. The camera body shall consist of at least two sections, a front and rear frame assembly connected by a bellows so that all seams are light-tight and dust-proof when the lens board is mounted in the front frame assembly, regardless of the angle to which the bed may be tilted.

3.3.1.1 Front frame assembly. The front frame assembly shall contain and hold a removable lens board in parallel alignment with the copyboard and film plane and shall also support the front section of the bellows. Adjustments shall be provided for raising and lowering the lens unit vertically and horizontally at least 1-1/2 inches (38mm) to each side of the center line of the frame assembly. The controls for the vertical and horizontal movement of the lens unit shall be located behind the rear frame assembly.

3.3.1.2 Rear frame assembly. The rear frame assembly shall provide a mounting for the removable camera backs and support for the rear section of the bellows. The camera back, when mounted to the rear frame assembly, shall hold the film holders in parallel alignment with the lensboard and copyboard. A vernier or fine focusing control shall be provided on the rear frame assembly for moving the focusing screen for critical focusing.

3.3.1.3 Bellows. The camera shall be provided with an extension bellows which will permit the front and rear frame assemblies to be moved to the limit of travel along the bed. The maximum bellows extension shall provide the capability of enlarging the 4 x 5 inch (102mm x 127mm) image with the longest focal lengths lens provided and the minimum extension shall allow the camera to focus at infinity with the shortest lens provided utilizing the 4 x 5 inch (102mm x 127mm) film holder. When fully extended, the bellows shall not interfere with the normal adjustment of the lens board nor shall it sag or collapse so as to interfere with the picture. It shall be made of a moisture and fungus resistant material and shall be light-tight with the camera back and lens board in any position.

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3.3.1.4 Camera back. The camera shall be provided with an easily removable, and replaceable, 360 degrees revolving back which accepts 8 inch by 10 inch (203mm by 254mm) film holders. A reducing back of the Graflok type which accepts 4 inch by 5 inch (102mm by 127mm) film holders shall also be provided. A focusing panel for each film holder shall be attached to the camera back by means of leaf springs which will permit the focusing panel to move outward in such a way that when the film holder is inserted, the film plane is positioned in the same plane as the focusing screen. The spring tension shall be sufficient to hold the film holder firmly and in its proper position in the focal plane of the camera and shall provide a light-tight seal.

3.3.1.4.1 Ground glass focusing screen. A ground glass focusing screen shall be provided as part of each focusing panel. The ground glass shall be mud ground, or equivalent. A clear one inch square section (non-ground) shall be provided in the center of the focusing screen to permit precision fine focusing. The glass screen shall be scored in 1/2 inch (12.5mm) square increments to provide for composing and aligning the image. One screen shall be 8 inches by 10 inches (203mm by 254mm) by 1/16 inch (1.6mm); the other screen shall be 4 inches by 5 inches (102mm by 127mm) by 1/16 inch (1.6mm).

3.3.1.5 Frame assemblies movement. The frame assemblies shall be capable of being traversed along the bed with a minimum of backlash. They shall be moved positively through a mechanical system which shall be actuated by a crank, or handwheel, located on the bed. Each frame assembly shall be provided with an adjustable friction device capable of applying constant braking effect along the line of contact with the bed and sufficient to hold the frame assembly firmly at any location regardless of the angle to which the bed may be tilted.

3.3.2 Camera bed. The camera bed shall provide a steady mounting for the camera body at any required location along the bed, or during its required travel thereon, and at any angle to which the camera and bed may be tilted. The bed shall be no more than 90 inches (2286mm) in length and shall be suitable for mounting the camera and its associated rack mechanisms and easel firmly thereon. The bed shall be equipped with a scale which will allow setting the lens (see 3.4) at the correct position for a desired enlargement or reduction without calculation or measurement.

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3.3.3 Easel assembly. The easel assembly shall be mounted permanently to the end of the camera bed facing the lens. It shall be of rigid construction and provide support for the lighting system. The easel shall be capable of alternately and conveniently accepting and holding the following:

Photographic masks (see 3.3.3.1)
Easel screens (see 3.3.3.2)

The easel shall be maintained in parallel alignment with respect to the lens board and film plane regardless of the angle to which the bed may be tilted.

3.3.3.1 Masks. A set of interlocking photographic masks consisting of the following sizes, shall be provided:

11 inches x 14 inches (279mm x 356mm)
8 inches x 10 inches (203mm x 254mm)
5 inches x 7 inches (127mm x 178mm)
4 inches x 5 inches (102mm x 127mm)
3-1/4 inches x 4 inches (82mm x 102mm)
2 inches x 2 inches (50mm x 50mm)

Each mask shall interlock with the one immediately larger and shall be held therein by suitable devices such as spring fingers. The 11 inches by 14 inches (279mm x 356mm) mask shall be suitably retained by the easel frame.

3.3.3.2 Screens. The following screens shall be provided for use with the easel: one glass screen, sand blasted on one side, and one translucent opal flashed glass screen for mounting transparencies for back lighting. The dimensions of the screens shall be 24 inches (610mm) by 24 inches (610mm) by 1/8 inch thick (3.2mm).

3.4 Lenses. The lenses provided with the camera set shall be a 12 inch (305mm) and a 6 inch (152mm) focal length lens and shall be type XII, semi-apochromat, copy lenses conforming to MIL-STD-150.

3.4.1 Relative aperture. The maximum relative aperture shall be f/9, or faster.

3.5 Shutter. The shutter shall be a behind-the-lens, solenoid operated electric shutter. The shutter speeds shall be from 1/125 second to 1 second plus B(bulb) and T(time).

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3.6 Cable release. A nominal 40 inches (1016mm) long, metal - or vinyl-sheathed cable release, threaded to fit the shutter's cable release socket, shall be provided.

3.7 Camera stand. The camera stand shall be an integral part of the camera set and shall have provisions for tilting the bed 90 degrees from the horizontal position. The base of the stand shall rest solidly on the floor. The stand shall provide adequate support and stability for the camera and bed assembly at any angular elevation of the camera bed, without the base being fastened to the floor. The stand shall include as an integral part an enclosed cabinet with shelves, doors, and door latches for storing the accessories used with the camera set.

3.8 Film holders. The following film holders, as specified in the contract or purchase order, shall be provided. The holders shall conform to MIL-H-3526 and the applicable size specified.

4 inches by 5 inches (102mm by 127mm) - Size 1
8 inches by 10 inches ((203mm by 254mm) - Size 2

3.9 Filter holder. A filter holder for 4-inch (102mm) square filter, adjustable to fit any lens barrel from 2-3/8 inches (60mm) to 3-7/8 inches (98mm) in diameter, shall be provided.

3.10 Lighting system. The lighting system shall consist of a pair of swinging arms with risers for supporting quartz type lamps. The system shall provide even illumination when used for illuminating, from the front, opaque materials and, from the back, transparent materials placed upon the easel assembly.

3.10.1 Swinging arms. The pair of swinging arms shall be fastened directly to the easel assembly and shall firmly support the two risers with quartz type lamps attached. The swinging arms shall position the lamps on each side of the easel assembly and shall be capable of pivoting the lamps to the front and to the rear of the easel assembly so as to provide even illumination on either side of the copy board. The swinging arms, with the lamps mounted on the risers, shall remain stable in any position required in the operation of the camera.

3.10.2 Lamp assemblies. The lamp assemblies shall be adjustable on the risers and shall remain stable in any position required in the operation of the camera. The lamps shall be of the quartz type, 650 watts nominal.

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3.10.2.1 Reflectors. Reflectors shall show no evidence of distortion, tarnishing, cracking, crazing, or loss of any reflecting characteristics after being subjected to the test specified in 4.12.

3.11 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touch up after mounting. The final paint film on type I surfaces shall be final film E, semigloss, light-gray enamel, conforming to MIL-F-14072.

3.12 Marking.

3.12.1 General. Marking shall conform to MIL-M-13231 (see 4.4). Each stand shall have a nameplate for the camera set.

3.12.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.12.3 Serial numbers. Each Camera Set, Still Picture KS-7( ) shall have a serial number.

3.13 Tropicalization.

3.13.1 Treatment of materials. Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then treated in accordance with the following:

3.13.2 Treating compounds. Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of treating compound shall be such that there shall be no increase in flammability of the treated material.

3.13.3 Toxicity. No treating materials shall be used that are identified in the Toxic Substance List (published by National Institute for Occupational Safety and Health) that will produce toxic effects via the respiratory tract, eye, skin (cutaneous and subcutaneous), or mouth. This includes handling the treated material during fabrication, transportation, operation or maintenance of the parts in which the treated materials are used or during use of the finished parts when used for the purpose intended. The contractor shall furnish certification data as specified on the contract that the treating materials used on the materials shall conform to the above.

3.13.4 Flexibility. Treatment shall not affect the flexibility of treated materials to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.13.5 Statement of treatment. The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use (see 6.2).

3.14 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.9).

3.15 Service conditions. The equipment shall meet the following service conditions:

3.15.1 Temperature.

3.15.1.1 Low. Before, during, and after testing as specified in 4.13.1, the equipment shall meet the following requirements.

Paragraph	Measurement	Performance
3.3 thru 3.10.2	Operation	Full specification requirement

3.15.1.2 High. Before, during, and after testing as specified in 4.13.2, the equipment shall meet the following requirements.

Paragraph	Measurement	Performance
3.3 thru 3.10.2	Operation	Full Specification requirement

3.15.2 Humidity. The Camera Set, Still Picture KS-7( ) shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials (see 4.7).

3.15.3 Altitude (elevation). The equipment shall meet full specification performance for the following measurement after testing as specified in 4.6.

Paragraph	Measurement
3.3 thru 3.10.2	Operation

3.16 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include preparation for delivery.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.15).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish two first article units of the complete Camera Set, Still Picture KS-7( ).

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

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Table I. First Article Inspection

Inspection	Req Para	Test Para	Unit 1	Unit 2
Inspection covered by subsidiary documents		4.4	Inspections to be performed on all units	
Group A inspection	See Table II		Inspections to be performed on all units	
Group B inspection	See Table III		Inspections to be performed on all units	
Environmental				
Low temperature	3.15.1.1	4.13.1	1	
High temperature	3.15.1.2	4.13.2	2	
Humidity	3.15.2	4.7		1
Altitude	3.15.3	4.6	3	

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Lenses	3.4
Film holders	3.8
Finish	3.11
Marking	3.12

4.5 Quality conformance inspection before preparation for delivery. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

Table II. Group A Inspection

Inspection	Req Para	Test Para	AQL	
			Major	Minor
Visual & Mechanical				
Camera	3.3 thru 3.3.3.2	4.14	1.5DPHU*	6.5DPHU
Shutter	3.5	4.14	1.0%	4.0%
Cable release	3.6	4.14	1.0%	4.0%
Camera stand	3.7	4.14	1.0%	4.0%
Filter holder	3.9	4.11	1.0%	4.0%
Lighting system	3.10	4.14	1.0%	4.0%
Camera set	3.16	4.14	1.0%	4.0%
Illumination	3.10	4.8	1.0%	**
Light-leak	3.3.1 thru 3.3.1.5	4.10	1.0%	**
Operational	3.3 thru 3.10.2	4.11	1.0%	**

*Defects per hundred units
**All operational defects are major

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

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4.5.2.1 Group B sampling plans. The Group B AQL shall be 4.0 percent defective and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

Table III. Group B Inspection

Inspection	Req Para	Test Para
Interchangeability	3.14	4.9
Reflectors	3.10.2.1	4.12

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

Table IV. Group C Inspection

Inspection	Req Para	Test Para
Altitude	3.15.3	4.6
Humidity	3.15.2	4.7
Temperature		
Low	3.15.1.1	4.13.1
High	3.15.1.2	4.13.2

4.5.3.1 Sampling for Group C inspection. One sample unit shall be selected at random for all inspections in Table IV from every 100 units or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in an order which is satisfactoy to the Government.

4.5.3.3 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

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4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Altitude (elevation). The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.15.3.

4.7 Humidity. The Camera Set, Still Picture, KS-7( ), shall be subjected to the humidity test specified in Method 507, Procedure II, of MIL-STD-810. Operation is not required during the humidity test. Inability of the camera set to meet the requirements of 3.15.2 shall constitute failure of the test.

4.8 Illumination. Illumination over the entire copyboard (easel) shall be measured using a photoelectric light meter, with a precision of ±5 percent, to determine compliance with the requirement specified in paragraph 3.10.

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions common to bed and front and rear frame assemblies and easel assembly, which govern mounting and alignment.

c. Dimensions of interlocking masks comprising the mask set.

d. Dimensions of easel screens.

e. Dimension of mounting plate and corresponding easel frame dimensions.

f. Location of hinges and fasteners on separable parts or assemblies which must mate.

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g. Dimensions of cavities which affect insertion of items.

4.10 Light-leak test. The camera shall be tested using the 8-inch x 10-inch (203mm x 254mm) and 4-inch x 5-inch (102mm by 127mm) film holders respectively to determine that it is light-tight to visible light. The camera loaded with 8-inch by 10-inch (203mm by 254mm) panchromatic film, ASA 125, with the dark slide withdrawn halfway and the bellows fully extended, shall have each side, top, front and back exposed to a light source, ASA Lamp Code EBV Photo lamp in a metal reflector, at a distance of 6 feet, for a period of 5 minutes. The slide shall be returned to the closed position and the film removed in a darkroom. After the film has been processed for normal exposure, examination shall show no evidence of exposure on the film. Repeat the operation with the 4-inch by 5-inch (102mm x 127mm) film and accessories.

4.11 Operational test. A minimum of six exposures with 4-inch by 5-inch (102mm x 127mm) and six exposures with 8-inch by 10-inch (203mm x 254mm) film shall be made. The lighting system shall be used as the light source. This test shall include taking exposures of both opaque and transparent materials. The camera and the bed shall be tilted and locked at 90 degrees from the horizontal. In this position, the camera bed, swinging arms and the lamp assemblies shall be rigidly and securely maintained in place; the bellows shall not sag or block out any portion of the image; the stand shall not move or tilt due to the unbalanced load, and no portion of the easel, front frame assembly, and rear frame assembly shall move or tilt from its fixed position.

4.12 Reflectors. The lighting system shall be operated intermittently for four hours on and one hour off for a total of 100 hours. Lamps shall be changed every twenty-five hours for this test. At the end of this test, the reflectors shall be inspected for conformance with the requirements of paragraph 3.10.2.1.

4.13 Temperature.

4.13.1 Low temperature. The equipment shall be tested in accordance with Method 502, Procedure I of MIL-STD-810. The storage temperature shall be -80 degrees F (-62 degrees C), not to exceed 24 hours, or to practical thermal equilibrium, which ever occurs first. The operating temperature shall be +50 degrees F (10 degrees C). Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.15.1.1.

4.13.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure II of MIL-STD-810. The operating temperature shall be +100 degrees F (38 degrees C). Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.15.1.2.

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4.14 Visual and mechanical inspection. Equipment shall be examined for the defects listed in MIL-STD-252 and table V.

Table V. Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Front frame assembly a. Misalignment b. Friction device does not brake properly or hold firmly. c. Traverse system binds, sticks, or freezes (coarse focusing). d. Vertical and horizontal lens adjustment binds, sticks, or freezes. e. Lens unit not accepted by camera. 2. Rear frame assembly a. Misalignment b. Friction device does not brake properly or hold firmly. c. Traverse system binds, sticks, or freezes (coarse focusing). d. Fine focusing control binds, sticks, or freezes. 3. Bellows a. When fully extended, sags or collapses so as to interfere with the picture. b. Loose fit c. Visual hole or crack 4. Camera back a. Not reversible to four positions b. Does not accept film holders. 5. Ground glass focusing screen a. Chipped, cracked or broken so as to prevent intended use. b. Scoring missing

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Table V. Classification of Visual and Mechanical Defects (cont)

Classification	Defects
Major	6. Camera bed a. Bed does not mount components firmly. b. Scale missing or defective. c. Bed alignment such that camera body movement binds, sticks, or freezes. 7. Easel Assembly a. Does not hold component parts properly. b. Misalignment 8. Mask set Missing, insufficient number, or wrong sizes. 9. Easel screens a. Chipped, cracked, or broken so as to prevent intended use. b. Wrong dimensions; will prevent intended use. 10. Calibrated mounting plate a. Wrong dimensions; will not properly fit into easel frame. b. Calibration missing, or not properly calibrated. 11. Camera stand a. Base does not properly support camera; out of alignment. b. Tilting improper or incomplete. c. Parts missing or broken.

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Table V. Classification of Visual and Mechanical Defects (cont)

Classification	Defects
Major	12. Lamp assembly a. Will not remain stable in position. b. Parts missing or so damaged as to prevent use. c. Reflectors bent or so damaged as to prevent intended use. d. Lamps broken or so damaged as to prevent insertion or use. 13. Swinging arm assembly a. Does not firmly support two banks of light. b. Does not pivot properly; binds or freezes. c. Does not remain stable in any position.
Minor	1. Front frame assembly Excessive backlash in traverse system 2. Rear frame assembly Excessive backlash in traverse system 3. Bellows a. Front or rear assembly does not move to full limit of travel along bed. b. When fully extended, interferes with normal adjustment of the lens board. c. Material not moisture or fungus resistant. 4. Camera back a. Not easily removed or replaced; sticks, binds or freezes. b. Focusing panel spring tension insufficient.

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Table V. Classification of Visual and Mechanical Defects (cont)

Classification	Defects
Minor	5. Ground glass focusing screen a. Slight chips or cracks; will not interfere with intended use. b. Scoring poorly defined, or non-uniform. 6. Easel assembly Loose mounting. 7. Mask set a. Masks do not interlock properly. b. Not suitably retained by easel frame. 8. Easel screens a. Slight chips or cracks; will not prevent intended use. b. Improperly finished. c. Wrong dimensions; will not prevent intended use. 9. Calibrated mounting plate a. Wrong dimensions; can be placed into easel frame. b. Calibration poorly defined or non-uniform. 10. Camera stand a. Doors do not fit properly. b. Latches do not hold properly.

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Table V. Classification of Visual and Mechanical Defects (cont)

Classification	Defects
Minor	11. Lamp assembly a. Loose of slightly damaged parts; will perform intended function. b. Cable assembly not strain relieved. c. Cable assembly has wrong dimensions. d. Cable assembly not permanently attached to lamp case. e. Slight scratches or minor damage to reflectors; will not affect intended use.

4.15 Acceptance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.16 Rough handling test (preparation for delivery). When rough handling test in accordance with MIL-P-116 is required by the contract (see 6.2c), the operational test of 4.11 shall be conducted after the rough handling test to determine freedom from operational malfunction caused by rough handling.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Camera Set, Still Picture KS-7( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Camera Set, Still Picture KS-7( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

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5.1.1.4.1 Technical literature and lens tissue. Each technical literature and lens tissue shall be packaged method IC-1.

5.1.1.4.2 Film holders and focusing backs. Each item shall be individually packaged method IC-1 as follows: Cushion each item by wrapping in paperboard conforming to PPP-P-291, type I. Secure the wrap with gummed paper tape conforming to PPP-T-45, type III, grade A. Place the cushioned items within a close-fitting bag conforming to MIL-B-117, type II, class C. Heat seal for closure.

5.1.1.4.3 Mounting plate, handwheels and ground glass. Each item shall be individually packaged method III as follows: Cushion each item by wrapping in paperboard conforming to PPP-P-291, type I. Secure the cushioning with gummed paper tape conforming to PPP-T-45, type III, grade A. Place the cushioned items within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.1.4.4 Lamp banks. Each lamp bank shall be individually packaged method III as follows: With lamps in place, fill all voids around each lamp with cellulose cushioning material conforming to PPP-C-843, type II, class B. Place a sheet of cellulose cushioning material on top of the bank of lamps. Place the cushioned lamp bank within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Box closure shall be in accordance with the appendix of the box specification.

5.1.1.4.5 Lens, shutter and cable release. Each item shall be individually packaged method IA-8 as follows: Cushion each item by wrapping in cellulose cushioning material conforming to PPP-C-843, type II, class B. Place the cushioned item within a close-fitting bag conforming to MIL-B-117, type II, class E. Heat seal for closure.

5.1.1.4.6 Cable assemblies. Each cable assembly shall be individually packaged in accordance with MIL-C-55442.

5.1.1.4.7 Carriage assembly. Each carriage assembly shall be individually packaged method III as follows: Remove the arms from each end of the assembly and the two crossbars. Any hardware removed shall be placed within a cloth drawstring bag conforming to PPP-B-20, and tied to the item from which removed. Nest the disassembled carriage assembly into a bundle of minimum dimensions. Cushion the assembly by wrapping in paperboard conforming to PPP-P-291, type I. Secure the cushioning with gummed paper tape conforming to PPP-T-45, type III, grade A. Place the cushioned carriage assembly within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Box closure shall be in accordance with the appendix of the box specification.

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5.1.1.4.8 Consolidated package. Consolidate the components, packaged as specified in 5.1.1.4.2 through 5.1.1.4.7, within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Box closure shall be in accordance with the appendix of the box specification.

5.1.1.4.9 Stand, camera. Each stand shall be individually packaged method III as follows: Place the consolidated package, as specified in 5.1.1.4.8 within the stand. Block and brace the consolidated package on the shelf, with dummy cells or pads of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, SW, grade W5c. Close the doors of the stand and secure the lock. Fabricate a wood cleated plywood box conforming to PPP-B-601, overseas type, style A. Place the stand on the skid base. Drill four holes through the skid base and the skid to correspond with the holes in the stand. The bolt holes shall be arranged so that the stand is centered on the skid base and when the box is fabricated, there is a two inch clearance between the stand, sides, ends and top of the box. Insert suitable size carriage bolts through the holes provided for in the skid base. The diameter of bolts shall be compatible with the diameter of the holes in the skid base. Place and secure a lock washer and nut over each bolt, and upset the threads protruding over the nut to prevent loosening. Place a waterproof shroud over the stand fabricated of barrier material conforming to PPP-B-1055, class H-5. No further packaging required.

5.1.1.4.10 Camera, copy, view type. Each camera shall be individually packaged method IA-16 as follows: Secure the camera on the bed in a close position. Fabricate a wood cleated plywood box conforming to PPP-B-601, overseas type, style A. The skid base shall be designed to accommodate the camera mounted on a wood cradle. Place the camera mounted in the wood cradle on the skid base. Drill four holes through the skid and cradle frame. The bolt holes shall be arranged so that the camera and cradle is centered on the skid base and when the box is fabricated, there is a two inch clearance between the camera, sides, ends and top of the box. Insert suitable size carriage bolts through the holes provided for in the skid base and cradle. Diameter of the bolts shall be compatible with the diameter of holes in the skid base and cradle. Assemble in order together with gaskets conforming to MIL-G-12803 over the mounting bolts. A pad of fiberboard conforming to PPP-F-320, type CF, class weather-resistant to extend beneath all points of contact between the skid base and cradle and a water vaporproof barrier conforming to MIL-B-131, class 1, and a shroud of paperboard conforming to PPP-P-291, type I. Waterproof around the bolts and gaskets at each application with adhesive conforming to MMM-A-1617, type 3. Place and secure a lock washer and nut over each bolt and upset the threads protruding over the nut to prevent loosening.

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Secure the camera within the cradle with wood blocking or metal strapping. Blocking material or metal strapping coming in contact with the guide rails shall be cushioned with cushioning material conforming to PPP-P-291, type I. Draw up the cushioning material over the camera and secure with gummed paper tape conforming to PPP-T-45, type III, grade A. Heat seal the water vaporproof barrier, leaving a small opening to exhaust the entrapped air. Exhaust the air and complete the seal. Secure the technical manual, packaged as specified in 5.1.1.4.1, on top of the barrier, with tape conforming to PPP-T-76.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2b).

5.2.1 Level A.

5.2.1.1 Stand, camera. The stand, packaged as specified in 5.1.1.4.9, shall be packed by assembling the remaining components of the box. Closure and strapping shall be in accordance with the appendix of the box specification.

5.2.1.2 Camera, copy, view type. The camera, unit packed as specified in 5.1.1.4.10, shall be packed by assembling the remaining components of the box. Closure and strapping shall be in accordance with the appendix of the box specification.

5.2.1.3 Each Camera Set, Still picture KS-7, packed as specified in 5.2.1, shall be arranged in the following sequence:

Box 1/2 Stand, camera
Box 2/2 Camera, copy, view type

5.2.2 Level B. Each Camera Set, Still Picture KS-7( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the boxes shall be domestic type.

5.2.3 Commercial packing. Packing shall be in accordance with FED STD No 356.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. Camera Set, Still Picture KS-7( ) is intended for use in copying photographs, line drawings and other flat copy; or for making enlargements, reductions, lantern slides and photographs of bulky objects, using 8-inch by 10-inch (203mm x 254mm) or 4-inch by 5-inch (102mm x 127mm) cut film.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or Commercial preservation and packing (see Section 5).

c. When rough handling test is required.

d. First article inspection (see 3.2 and 4.3).

e. Submission of the statement of treatment referenced in 3.13.5, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

f. Group C inspection failure procedure (see 4.5.3.3).

g. If Group B and Group C sample units are not to be accepted on contract (see 4.5.4).

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: KS-7W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specification or preparing activity for recommended disposability methods.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-13959B(EL) CAMERA SET, STILL PICTURE KS-7( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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